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Strangers to the Stars: Abstraction, Aeriality, Aspect Perception

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'Give me truths', wrote Ralph Waldo Emerson in his poem 'Blight' (1847), 'For I am weary of the surfaces / And die of inanition' (1994: 111). The poem is a scalding critique of the instrumentalisation of a natural world invaded 'impiously for gain' and the alienating consequences of this pinched yet conquering rationalism. For Emerson, as for Paul Virilio, the eye has taken on 'the function of a weapon' (Virilio 1989: 3) with the power to capture and command, yet such authority is not without cost, as Emerson's assessment of the impoverishment that comes with full spectrum dominance suggests: 'Our eyes / Are armed, but we are strangers to the stars' (1994: 112). The act of looking here is an act of aggressive appropriation that, in opposition to commonplace assumptions about seeing and believing, does not yield truth but only a debilitating emptiness. What is perceived, Virilio concludes, in terms of a military optics Emerson would have recognised, is already lost (1989: 4).

Eleven years after the publication of 'Blight', Nadar made the first aerial photograph from a balloon over Paris; two years later, in 1860, James Wallace Black produced the first aerial photograph in the US, 1,200 feet above the streets of Emerson's Boston. While aerial reconnaissance from balloons was first used during the American Civil War, it was with World War I that aerial photography for military purposes properly emerged, the vertical aerial image instantiating the optical domination of the flattened and estranged surface world anticipated in 'Blight'. Reading aerial reconnaissance photographs required highly trained interpreters who, like Emerson's 'young scholars, who invade our hills, / Bold as the engineer who fells the wood / And travelling often in the cut he makes' (1994: 111), parse representations of an environment that has been collapsed and denatured for them alone. The closed

loop created here between mode of representation and its legibility effectively, as Bernd Hüppauf argues, 'kill[s] the natural landscape and replace[s] it with highly artificial and, within its own parameters, functional spatial arrangements' (1995: 6). In Donna Haraway's pithy formulation this assault by the God's-eye view 'fucks the world to make techno-monsters' (1991: 189).

By eliminating the horizon the aerial photographic view produces an all-over visual field that, without supplementary information or the training required to travel in the cut made by the camera that felled the wood, is primarily apprehended as opaque form. In this way, aerial photographs can seem as distant and blankly non-figurative as abstract painting, as if the vertical vantage that makes the aerial view possible has collapsed and left just the flat plane of the image. While the commanding function of the aerial view, now as in World War I, is commonly used by state agencies for purposes of surveillance, control, targeting and destruction, aerial photographs are also exhibited as works of art. From Emil Schulthess, Mario Giacomelli, William Garnett and Ed Ruscha to Emmet Gowin, David Maisel, David T. Hanson and Terry Evans, aerial photography has become an established genre of art photography (Cosgrove and Fox 2010: 99–138). Beyond the functional purpose of data gathering, though, what cultural and aesthetic work might aerial photographs of this kind be expected to do? Often aerial art photography is directly engaged in a self-reflexive examination of the promise and limitations of the vertical perspective as it has been used as a mode of capture and control. Much contemporary aerial photography is concerned with issues of industrial despoliation and military concealment, with strategies of visibility and invisibility, display and concealment, camouflage and surveillance.

What is also the case, though, is that in approaching the complex history of vertical imaging as a mode of domination, aerial art photography often appears to reproduce the obfuscatory strategies it seeks to challenge. A large part of the difficulty here lies in the meaning and purpose of abstraction and its apparent resistance to interpretation. In engaging with the abstracting power of the aerial view, recent photography draws attention to the ways in which abstract form has been and can be used as a form of ideological camouflage that screens from view content that is considered best hidden, such as contaminated military test sites and large-scale industrial devastation of forests and mountains.

At the same time, this critical engagement also deploys abstraction in order to interrogate its function, a kind of counter-surveillance that replays camouflaging strategies as a form of exposure, a form of showing rather than hiding. Nevertheless, the abstract image's ambiguity and indeterminacy does not make this intervention especially obvious. The slippery nature of abstraction, especially with regard to the distinctions and similarities to be made between forms of abstract visual art (painting and photography) and in terms of the conventional opposition between surface and depth, requires serious consideration in order to establish the possibility of an aerial view that might move beyond the ultimately unhelpful binary deployed by Emerson. The weariness toward 'the surfaces' announced in 'Blight', with its denigration of surface as falsehood in favour of the truth of depth, is, I want to argue here, misplaced and in need of revision. What if whatever truths there may be are in fact embodied in the surface itself? Can the hidden be a form of display? Does every act of concealment actually call attention to itself? These are some of the questions, I think, raised by aerial art photography and directed toward the history of abstraction in twentieth-century visual art and its links to the military–industrial optic of capture and command.

Is There Life on Earth?

In a 1972 article for *Engineering and Science*, in order to explain what the Mariner 9 spacecraft needed to do in order to detect possible signs of life on Mars, the astronomer Carl Sagan turned the tables and imagined how close a Martian observer would need to be in order to detect life on this planet. Beginning with thousands of images produced from weather satellites which photograph the Earth at one-kilometre resolution, Sagan found the pictures to be 'biologically uninteresting'. At one-kilometre resolution, he concludes, 'there is no sign of life – intelligent or otherwise – in Washington, Boston, New York City, Moscow, Peking, Melbourne, Berlin, Paris, London, or any other major population center' (Sagan 1972: 17). Zooming in to examine 1,800 high-resolution colour photographs produced by Apollo and Gemini astronauts at 100-metre resolution, Sagan detects some evidence of life, but only in a mere 57 images. Equipped with a knowledge of the Earth's features, it is possible to identify roads (29), canals (5), agricultural geometricising (15), jet contrails (4) and industrial

pollution (4). Cities with large highway systems like Dallas–Fort Worth are easily detectable, but large cities without such systems are not. At 100 metres, then, it is possible to detect evidence of human life, but not much; beyond a kilometre the earth yields no evidence of habitation at all. Life that does not rework the environment (such as animals and trees) is even harder to see and only detectable at one- to ten-metre resolutions (Sagan 1972: 19).

In an earlier draft of these findings, Sagan concludes from the evidence that '[n]ature paints with a broad brush, but Man, with a culture far from global, is a pointillist' (Sagan and Wallace 1970: 5). It is a throwaway line, disposable enough not to make it into the published version, but Sagan's choice of metaphor is both familiar and revealing. The distance provided by the aerial perspective tends to render phenomena invisible more often than it reveals and instead what is shown are formal configurations that bring to mind the shapes and patterns of modernist painting. Refusing to yield the kind of information that close and expert interpretation of photographic data ought to uncover, the image collapses into the surface of a painterly abstraction.

Fear of Abstraction

The tendency to conflate the aerial view and modernist painting is in no small measure a consequence of the aeroplane and Cubism (especially) appearing at around the same time (Saint-Amour 2003: 349). From Gertrude Stein to John Piper, the temptation to liken the view from above to the paintings of Miró or Picasso has been irresistible (Hauser 2007: 189). Writing in 1984, however, the cultural geographer J. B. Jackson complained that it had become a lazy habit, when looking at 'large-scale organizations of man-made spaces' from the air, to compare the view of the land to familiar patterns, tapestries, floor coverings, 'or to the work of some painter: Mondrian or Fernand Léger or Diebenkorn' (1984: 141). A more detailed study of the surface of the aerial view, Jackson argues, should reveal rather than conceal what lies beneath. While it is understandable that a geographer expects to utilise photographs for data gathering rather than for aesthetic contemplation, Jackson's dismissal of the lazy habit of seeing the aerial view as if it were art is revealing of a visual sense that responds to surface primarily as form. To see abstract images as somehow in the way of the truth of the landscape says something

about the powerful suspicion of surface as a veil behind which 'real' meaning hides. Is it really possible to block out the formal properties of the surface – as enemy of truth – while the buried significance is excavated by close reading? Refusal to trust the surface appearance of the image as evidence can also belie the otherwise persistent faith in photographic representation as the truth made visible: consider the unshakable misconception during the Cold War that the Soviets had masses of ICBMs hidden somewhere where aerial reconnaissance could not find them, or the more recent puzzle of Saddam Hussein's weapons of mass destruction. The paradox here is that the evidence produced by a surveillance system based on the purported 'truth' of photography is refused if it is not consistent with need.

A similar discomfort with abstraction can be seen in environmental writer Wendell Berry's preface to a volume of David T. Hanson's aerial photographs of US military–industrial despoliation, *Waste Land: Meditations on a Ravaged Landscape* (1997). Berry expects 'that some people will account for these photographic images as "abstract art," or will see them as "beautiful shapes"'. A closer look, though, Berry argues, will show that 'nothing in them is abstract and that their common subject is a monstrous ugliness' (Berry in Hanson 1997: 3). Berry is clearly resisting the optical immediacy of the image in favour of its informational, documentary, readerly function and to the extent that the image is offered as evidence of the usually unseen management of space, this is an appropriate warning against the aestheticisation and neutralisation of a consciously political form of representation. Indeed, the fact that Hanson's images are accompanied with captions and information on the sites depicted surely does suggest that the photographer is determined to avoid the absorption of the photographs into an unreflective discourse of visual beauty. Yet at the same time Berry is surely fighting a losing battle against the visual medium: form is apprehended first, everything else follows. Hanson himself explains that the aerial view 'seemed the most appropriate form of representation for the late twentieth-century landscape: an abstracted and distanced technological view of the earth, mirroring the military's applications of aerial photography for surveillance and targeting'. The aerial view, he goes on:

> realizes the Cartesian rationalization and abstraction of space that has preoccupied Western culture and visual art for the last three hundred

> years [and] also allows for the framing of relationships between objects that may seem unrelated on the ground, and it permits access to sites with security restrictions. What otherwise cannot be pictured becomes available to the camera. (Hanson 1997: 5)

As compelling as Hanson's explanation of his practice is here, the problem is that it still relies on the negative associations of the abstract aerial view as the unequivocal sign of power. What Jackson, Berry and other commentators uncomfortable with the aerial image's capacity to awaken unreflective aesthetic responses are sceptical of is the power of the 'God's-eye view' to dissolve all critique into a diverting spectacle of abstraction. This is certainly the common reading of the emergence of aerial photography as a decisive technology during World War I, evident in Hüppauf's assessment of industrialised war 'kill[ing] the natural landscape' and replacing it with 'functional spatial arrangements'. Aerial photographs, Hüppauf continues, 'resemble abstract landscape paintings, which also reduce the profusion of details of a "natural" landscape to a "rationally" structured order' (1995: 6). Interpreting the vertical perspective as the de facto position of power in this way, Paul K. Saint-Amour argues, has meant that the horizontal view has been oppositionally situated as the subordinated site within which resistance might arise: 'the vertical is the axis of order, paradigm, symbolic function, disutility, unimpeded sightlines, and disembodied omniscience; whereas to the horizontal belong disorder, syntagm, enunciative function, utility, partial sightlines, and exposure to visibility' (2011: 246). The problem with this binary interpretation, for Saint-Amour, is that it does not allow room for 'differential qualities' within the abstract space of the aerial view itself because they have 'long been camouflaged, ignored, or misattributed exclusively to the ground' (248).

No small part of the hostility toward seeing aerial photographs as analogues for abstract painting, then, is due to the perceived impenetrability of the abstract surface as a display of authoritarian power. In fact, abstraction is more complicated than that, as Saint-Amour suggests. During World War I the photographer Edward Steichen became commander of the photographic division of the American Expeditionary Forces, his experience of aerial photography on the Western Front leading him to renounce the pictorialism of which he had been, until this point, a leading exponent. Instead Steichen became a powerful advocate of photography's

capacity for sharp, unretouched realism, a move that, for Virilio, is paradigmatic of the way in which the technologies of photography and war converged in the twentieth century. This convergence transformed the battlefield and the relationship between war and art: 'The image is no longer solitary (subjective, elitist, artisanal); it is solidary (objective, democratic, industrial). There is no longer a unique image as in art, but the manufacture of countless prints, a vast panoply of imagery synthetically reproducing the natural restlessness of the spectator's eye' (Virilio 1994: 53). While it seems that the fact of photography's integration into the optics of twentieth-century industrialised warfare compelled Steichen to renounce art photography, what is revealing is Virilio's observation that 'Steichen claimed to have only been able to carry out his military mission properly thanks to his knowledge of French art (the Impressionists, the Cubists and especially the work of Rodin)' (49). It is this fusion of art and military–industrial production that comes to shape Steichen's understanding of photography, an awkward convergence marked by the fact that Steichen held on to thousands of wartime images he did not himself take yet 'exhibited and sold as products of Steichen's authorship and as his property' (Virilio 1994: 53; Sekula 1975). Transposed into the field of art, wartime aerial photography levers open some of the 'differential qualities' latent in the abstract view, even if, in Steichen's case, a lingering desire to hang on to the aura of the creative individual and the unique art object gets in the way. The resistant properties of modernist abstraction and the mechanically produced aerial view do, I think, suggest ways of thinking about the vertical perspective and its collapse of the third dimension into surface that are not reductively instrumental and rationalistic. To see how this might work involves a brush with high formalism.

Forced to the Surface

The American art critic Clement Greenberg argued in his influential early essay 'Towards a Newer Laocoon' that in order for modern painting to claim its position as an art it had to 'surrender to the resistance of its medium; which resistance consists chiefly in the flat picture plane's denial of efforts to "hole through" it for realistic perspectival space' (Greenberg 2000 [1940]: 68). For Greenberg, yielding to the medium meant that painting could finally become itself and be no longer a vehicle, subordinate to

illusion, narrative or any extraneous function of what Greenberg broadly terms 'literature'. Acceptance of the flatness of the picture plane and the rectangular shape of the canvas 'got rid of imitation' (68) and the apparatus of an illusionistic third dimension in favour of an interrogation of the medium as itself: brushstroke, colour, line, shape, surface. As Greenberg tells it, painting's process of resistance and surrender is one of dynamic struggle where the medium strains and pushes in order to realise itself. This is not a story of artists or movements; instead, Greenberg provides a narrative of force and counterforce, of pressure and volume and weight. In a remarkable passage, Greenberg writes that 'the picture plane itself grows shallower and shallower, flattening out and pressing together the fictive planes of depth until they meet as one upon the real and material plane which is the actual surface of the canvas; where they lie side by side or interlocked or transparently imposed upon each other' (68). This is a plastic, almost geophysical account of tectonic shift and realignment where 'real and material' things are active forces working upon one another in space and time. Under these conditions, if the painter does try to represent objects, 'their shapes flatten and spread in the dense, two-dimensional atmosphere' (68). Faced with the pressure exerted by the medium, a 'vibrating tension is set up as the objects struggle to maintain their volume against the tendency of the real picture plane to re-assert its material flatness and crush them to silhouettes' (68). As a result of this tension, like some seismic release, 'realistic space cracks and splinters into flat planes which come forward, parallel to the plane surface' (68). This push to the surface, Greenberg notes, can be accelerated by a violent puncturing of illusionistic space made by the insertion of found materials (newspaper, wood) which 'destroy the partial illusion of depth by slamming the various planes together' (68). While the artist here can push things along, the direction of events is driven by some kind of physical inexorability. In Greenberg's account, the annihilation of illusionist space is virtually apocalyptic – volcanic, catastrophic, purgative – and results in the perfect alignment of the elements: 'the pristine flatness of the stretched canvas' (68). With the universe now in order, art can get on with the business of being art.

By describing the work of the modernist avant-garde as a process of tectonic resistance and release, Greenberg positions the shifting plates of art as autonomous from individual artists, tendencies or

movements. The ground of art seems to be moving all by itself, like an indifferent earth held together only by the immutability of certain physical laws. The emergence of abstraction is cataclysmic, like an earthquake or tsunami, pressing, cracking, splintering and slamming until the illusion of depth is shattered and dispersed. Imagining the extinction of perspectival space to be a consequence of the internal logic of painting – the force of the medium ultimately becoming too great to resist – allows Greenberg to position the triumph of flatness as the inevitable outcome of painting's interrogation of its own limits. Yet the death of the dinosaur is not a geophysical inevitability; as T. J. Clark points out, flatness is compelling not in and of itself but because it stands in critical relation to perspectival depth, both epistemologically (flatness 'standing for the truth of *seeing*, the actual form of our knowledge of things') and politically. As an aggressive challenge to bourgeois consumption, according to Clark, 'flatness appeared as a barrier to the ordinary bourgeois' wish to enter a picture and dream, to have it be a space apart from life in which the mind would be free to make its own connections' (1982: 152). Here, flatness is not formally inevitable but it does, as in Greenberg's formulation, stand as a material, physical barrier to subjectivist reverie: art is not an escape from the world but a confrontation with it.

The tectonic collisions in Greenberg's field of art conjure a dynamic arena somehow beyond human agency and certainly beyond human scale; this is art operating in spaces inhospitable to ordinary representation. Indeed, what is most effective about Greenberg's account of how illusion surrenders to the medium of painting is the way that, like abstract painting, the description creates an entirely non-figurative space. Greenberg's narrative provides no space in which objects could be plausibly situated; it is a field of action – of planes, extensions, force – beyond any conventional system of representation. In a later essay on post-war American abstraction, Greenberg explains that such painting has abandoned 'the representation of the kind of space that recognizable, three-dimensional objects can inhabit' (Greenberg quoted in Harrison 1993: 201). This, I think, is precisely what he is after in 'Towards a Newer Laocoon': a kind of primordial space that no one can enter and dream, upon which no fantasies can be projected, where only the imperatives of the medium itself can be said to have authority and substance. This space of abstraction is like a world before and after history, a space of refusal so profound

it has voided every trace of where it might come from and what might have motivated its creation and instead appears simply as brute matter, beyond all comment.

Greenberg's story of the coming of abstraction is deliberately non-figurative in the same way that abstract painting is non-figurative; the purpose of this non-figuration lies in its relative and negative power. As Charles Harrison explains, non-figurativeness is always relative 'both to the expectations aroused by antecedent and contemporary figurative art and, once a tradition of abstract art has become established, to the extent of negation of those expectations which has so far been reached within that tradition' (1993: 202–3). If the non-figurative presupposes that the figurative is what is normally expected, the radicalism of Greenberg's description of the emergence of flatness lies in its defamiliarisation of the narrative of art history: Greenberg's non-figuration negates the conventional story of art and, to recall Clark's point about flatness, attempts to embody 'the actual form of our knowledge of things' (1982: 152). What is important here is the pressure Greenberg – and the abstract painting he describes – puts upon expectations about art, what it is for and what it can show, not to mention the simultaneous pressure exerted upon thinking about art and what that is for. The negation of perspectival space is profound and driven by, as in Clark's reading, a critical refusal of the notion of art as 'a space apart from life'. Abstraction is oppositional and instead of dreams and illusions it offers the unadorned fact of its physicality as surface.

In order to activate its own critical position, the work of abstract art, for Harrison, 'must first evoke and set in play those very functions it aims to disparage' (1993: 203). To be considered as painting the work somehow has to do what paintings are expected to do but at the same time reject those expectations. In other words, to make clear its criticality, the abstract work must call attention to what it is not; it must recall the production of illusionistic depth even as it denies this. Greenberg accepted that '[t]he first mark made on a surface destroys its virtual flatness' (1982: 8); the relationship between the mark and the surface is a kind of figure-and-ground relationship, but one that is doubly there and not there. As Harrison writes: 'An abstract painting is something standing in place of a picture, which is nevertheless not a picture *of* anything' (1993: 203).

In light of this discussion, Sagan's leap from the aerial photograph

to the abstract painting may seem more irrelevant than ever, a lazy comment based solely on no more than a broad resemblance. While the horizonless, pattern-dominated aerial photograph replicates the all-over surface of the abstract painting, the two seemingly have little else in common ontologically or aesthetically. Certainly, because of their expressed interest in flying there is a clear relationship between the work of abstract painters like Peter Lanyon and Richard Diebenkorn and the aerial view, and between aerial images by photographers like William Garnett and modernist abstraction. The nature of that relationship however, may be no more than formal in the loosest sense. Yet the critical discourse surrounding modernist abstraction still provides a means of interrogating the function of abstraction in aerial photography and its relationship to issues of visibility, invisibility and power. What the art historical analysis of modernist abstraction offers is a way of thinking about forms of art as in complex, often antagonistic relationships with each other; a way of thinking about the non-figurative that holds the formal and political in critical tension; and a means of approaching the implications of surface and depth.

Harrison's point about abstract painting is that one form of art is necessarily interpreted through its relationship to other forms. Reading aerial photographs as abstract paintings does not have to collapse the two but can allow one form to be read through the other; the abstraction of the aerial photograph appears to deliver what we expect from abstract painting but does not. Greenberg notes the doubleness in the painting's pristine surface that is nevertheless interrupted by the mark; the aerial photograph is also double in the sense that it may superficially appear to be like an abstract painting but its surface carries no marks. In this way the abstract photograph works in the opposite direction to Greenberg's painting. The absence of the mark confirms the flatness of the surface of the image; the photograph resembles a painting but is not. Yet what the aerial photograph smuggles back into the equation is precisely what the abstract painting has denied: the plausible representation of a space occupied by three-dimensional objects. At the same time, by resembling the abstract painting, the aerial photograph defamiliarises three-dimensional space enough to preserve the element of negation abstract art opened up.

Greenberg's almost non-figurative writing about abstraction as a field of contending forces gets close to achieving the paradoxical condition of a non-anthropomorphic reading of culture much as

Steichen came to be drawn to photography's hard-edged resistance to human intervention (photographs, as Roland Barthes reminds us, seem to belong to that class of images 'not made by the hand of man' [1981: 82]). The alienation effect Greenberg manages to generate here, where human agency appears as non-human physical force, is an effect often achieved in aerial photographs where distance renders evidence of human activity as indistinguishable from geophysical or biological process. Although this alienating impenetrable surface is so often the target of hostility toward instrumental rationality's classification and domination of the object world – the source of Emerson's weariness – it is here also the means of blocking the penetrating assault by perspectival rationalism. Greenberg insists on the imperatives of form as the material signs of struggle; what the aerial photograph opens up is a sense of epistemological doubt about the nature of the forms depicted in the images. Nevertheless, it is through form that the aerial photograph announces its concerns and presents its evidence. The abstract painting and the aerial photograph both raise complex questions regarding the meaning and value of surface and depth. If the surface of the abstract painting rebuffs illusionistic perspectival depth, the pristine surface of the photograph confirms the material flatness of the picture plane even as every millimetre of that surface calls out for interpretive reconstruction of a lost third dimension.

Abstraction and Homeland

On 18 May 1980, Mount St Helens, one of several volcanic peaks that dominate the Cascade Range of the Pacific Northwest, exploded after two months of intense earthquake activity and a series of weak, intermittent eruptions. In the Washington area at the time, having been awarded a National Endowment for the Arts fellowship to undertake a photographic project in the state, was Princeton-based photographer Emmet Gowin. Intent on documenting the devastation, security restrictions meant that the best way to approach the site was from the air (Reynolds 2002: 143). While Frederick Sommer's horizonless landscapes were already a big influence on Gowin's work, he had made no aerial photographs until this point, yet the Mount St Helens project would result in Gowin becoming one of the most prominent aerial photographers working in the US, and he would return to the site

of the earthquake over a number of years. The Mount St Helens explosion also drew to the area landscape photographer Frank Gohlke who, like Gowin, returned numerous times through the 1980s. The scale of the disaster – 40,000 acres levelled, 20,000 additional acres of timber damaged, more than 60 people killed – meant that, as Gohlke wrote in 1985, 'Mount St. Helens is the only place on the continent where one can *see* so clearly the effects of forces comparable in scale to those produced by nuclear weapons' (2009 [1985]: 97). Gohlke's move from natural to nuclear catastrophe here perhaps reflects the political climate of the mid-1980s during President Ronald Reagan's revivified Cold War. The fear of and fascination with destruction that Gohlke admits drew him to Mount St Helens, though, is also about the American West's historical place as the double site of US natural and technological sublime.

This convergence of geophysical and historical sources of awe and dread identified by Gohlke is also not far away during Gowin's explorations of Mount St Helens. A chance flight over the Hanford Reservation in 1986, Gowin notes, 'changed my whole perception of the age in which we live' (Gowin in Reynolds 2002: 144). Originally built as part of the Manhattan Project, and home to the first full-scale plutonium reactor in the world, during the Cold War Hanford in south-central Washington expanded to nine nuclear reactors and five large plutonium processing complexes. Plutonium for most of the US nuclear arsenal was produced at Hanford, and though the site was decommissioned at the end of the Cold War it remains the most contaminated place in the United States. What emerged for Gowin in his flight over Hanford was 'a pattern of relationships and a dark history of places and events':

> Still visible after forty years were the pathways, burial mounds, and waste disposal trenches, as well as skeletal remains of a city once used by over thirty thousand people who built the first reactors and enriched the first uranium. Etched and carved into the body of the desert landscape below was a whole history of unconscious traces . . . What I saw, imagined, and now know, was that a landscape had been created that could never be saved. (Gowin in Reynolds 2002: 144)

From Hanford, which from the air exposes its history as a series of ghostly scratches and scars, Gowin began to seek out other nuclear sites, 'realities that [he] had unconsciously forgotten' (144). The

power of the aerial view for Gowin here resides, as it does for Hanson, in its capacity to expose the signs simply not available to the ground-level observer, a point emphasised by Kitty Hauser in her discussion of aerial archaeology: 'once soil has been disturbed . . . that disturbance is essentially virtually ineradicable, no matter how long ago it took place' (2007: 163). Nonetheless, without the supporting paraphernalia of captions, maps or other contextualising information, such as Gowin's explanation of his investigative surveillance of sites from the air, the images are still in danger of looking like anything at all.

Discussion of aerial photographs, as we have seen in Jackson and Berry, invariably turns to this dilemma. Discussing Gowin's images of Western US military sites, environmental writer Terry Tempest Williams asks:

> Do we view each photograph as a work of art and simply enjoy the pleasure of form and texture as its own aesthetic statement? Earth becomes the artist's canvas. Or do we see this portfolio of images as a haunting documentation of place, how we have altered the land with our industrial and technological might from agriculture to recreation to national security? (2002: 126)

The answer for Williams is both; Gowin shows the earth as 'abstraction and homeland' (126). Peter Goin, another photographer of nuclear landscapes (though not aerial views), admits that '[f]ormal beauty can be a contradictory element in a photograph that comments critically on land use and land management' (1991: xxii), but it is this contradictory element that saves aerial photographs like Hanson's and Gowin's from functioning merely as decoration, since their abstract surface aligns them with the confrontational energy of modernist painting. Hanson's and Gowin's images, while carrying the awful toxic information Berry wants us to carefully see, present themselves as pure form. While this form may look like aestheticist acquiescence, it is the abstract form itself that refuses to allow us to reductively convert the photographs back into pure information as some kind of objective truth. As form the images stand apart from the instrumentality of vision expected of the camera as part of the apparatus of surveillance and disciplinary intelligence. What Berry wants us to look at we cannot see, and what he does not want us to dwell on is precisely what we do see.

Williams's identification of a doubleness in Gowin's work is important here and the terms she uses to describe that duality are also revealing, the coupling of abstraction and homeland bringing together aesthetic form and politics ('homeland' post-9/11 is an especially freighted word) as aspects of each other. What Williams is suggesting, I think, is a way of thinking about doubling as a form of simultaneous showing and hiding that speaks both to the ambiguities of abstract form evident in aerial art photographs and to the politics of surveillance and security embedded in the vertical view. Commentators who want to shut down an aesthetic response to aerial photography because it somehow occludes or dismisses any critical clout the images may carry have missed, if Williams is right, the power of such images to embody the contradictory phenomenon of a power that hides by showing itself (homeland as abstraction) and where any attempt to expose that power ends up calling attention only to itself (abstraction as homeland).

The doubleness Williams identifies in Gowin's aerial shots recalls, to return briefly to painting, a suggestive discussion by art historian Fred Orton of Jasper Johns's famous mid-1950s painting of the Stars and Stripes entitled *Flag*. What is important about Johns's work, I think, is that it stands between the poles of Greenbergian abstract formalism and the violent return of figuration in Pop Art. It is also, of course, a painting with something to say about form and politics, though exactly what is hard to read. *Flag*, then, seems to be operating in two directions at the same time in ways that are not dissimilar, at least in Orton's reading, to what I believe is going on in aerial art photography. Orton argues that the painting works by transmitting two main messages simultaneously, one about art and one about flags. For Orton, '[a]s a work of art [the painting] embodies a set of ideas and beliefs about art and aesthetics and as the American flag it embodies a set of ideas and beliefs about citizenship, nationalism and patriotism' (1994: 140). Any attempt to concentrate on the painting's 'message as a work of art', Orton continues, is 'interrupted' by its message as an American flag:

> This interruption causes doubts about *Flag*'s ontology as a work of art but not about its patrioticness. Rather, this interruption confirms the work of art's patriotic value . . . But when we try to concentrate on *Flag*'s message as the American flag, the patriotic message is interrupted by the art message . . . The meaning of *Flag*'s surface in art

> must somehow be at odds – this is the character of an interruption – with its meaning as the American flag if it is not to be understood – and never has been understood in this way – purely and simply as the American flag. (1994: 140–1)

Read this way, *Flag* might be understood as functioning as a form of dazzle painting or camouflage, its surface interfering with the signals it is itself simultaneously sending. Like Wittgenstein's duck–rabbit illustration, which is used to explore the differences between seeing an image *of* a duck or rabbit ('passive' seeing) and seeing the image *as* a duck or a rabbit (interpretive seeing), seeing a flag is distinct from seeing the painting as a flag (see Wittgenstein 1968: 194–7; Bull 1999: 21–4). And, like Wittgenstein's example, *Flag* carries more than one message at once – a message as art and about flags; it is both a duck and a rabbit, and to see both requires an oscillation between the two. The two readings, as Orton suggests, do not sit comfortably together; rather, one is always interrupting the other. This is the abstraction/homeland oscillation Williams identifies in Gowin's work and which, I think, is crucial to an interpretation of aerial art photography that does not dismiss consideration of abstraction as a reactionary formalism but instead reads it as intrinsic to what the images can show and not show about power and vision.

Aspect-Dawning

The stubborn foregrounding of abstract form is most obviously present in the work of David Maisel, a student of Gowin's at Princeton and his assistant on the Mount St Helens project. After working with Gowin, Maisel went on to photograph the logging industry of northern Maine; open pit mines in the West, first in black and white, then in lurid colour; Owens Lake in Los Angeles; the Great Salt Lake in Utah; and, in negative, aerial views of LA. Maisel's work revels in the contradiction between abstract form and the overdetermined site already noted in work like Hanson's and Gowin's. The photographs from the 2005 series *Terminal Mirage* depicting the area around the Great Salt Lake in Utah, for example, supply no contextualising support beyond the title. The Great Salt Lake is one of the biggest terminal lakes in the world; it is also in the middle of the desert where mirages are common. So the title is prosaically factual at one level but also invokes the

entropic sci-fi scenarios of J. G. Ballard and Robert Smithson. *Terminal Mirage* calls up an entire repertoire of tropes relating to disease and decay, duplicity and illusion, but beyond these two words, the images are left to fend for themselves.

The scale of Maisel's images is hard to discern; the square format and absence of horizon resist a reading of the surface as landscape and the colours are implausibly vivid. There must be something wrong when bodies of liquid are angry reds, purples and greens, but it is impossible to tell from the images what this chromatic excess signifies (Figure 4.1). The Great Salt Lake, as well as being terminal, is naturally rich in five major elements, each with its own colour, which seep into the water and are concentrated in the lake. Other sources of colour in the lake are species of algae and bacteria; industrial evaporation ponds for extracting salts and minerals; toxic chemical weapons waste; and contaminated wastewater ponds of the Magnesium Corporation of America. This information expands but does not resolve the information in the photograph, which sends simultaneous messages that cannot be comfortably received at the same time: it is a photograph that looks like a painting; it is a photograph of a real place; within that real place are colours that look like paint; some of those colours are natural, some are manufactured. To recall Orton, the image is sending messages as documentary and messages as art, one interrupting the other. The meaning of the photograph's surface as form gets in the way of its documentary function as a picture of the Great Salt Lake; the art message calls up an implied context within the visual history of abstract forms even as the image reports on the garish deformations wrought by natural and unnatural processes on the ground. Donna Haraway's point that the vertical view 'fucks the world to make techno-monsters' must also be modified or expanded here to include the fact that the techno-monsters have already fucked the world so that Maisel can take these photographs.

William Fox has observed that Maisel's images 'can be abstract in the extreme and run exactly counter to the intent of most aerial photography, which is to specify and identify pattern and activity in the landscape' (2009: 116). The absence of grounding information in Maisel's work may be, as Fox suggests, because 'people used to spend more time in front of his work discussing pollution than appreciating the photographs as artworks' (116), and it is probably true that if the primary function of the photographs was

to deliver compelling evidence of environmental negligence then intoxicating abstractions are unlikely to work as documentation. What I think Maisel knows is that aerial abstraction is less effective as a direct truth-telling device and more about an intense oscillation between showing and hiding and the resistant and seductive properties of surface. What the Salt Lake photographs show is a mess of natural and unnatural elements, a scrambled set of codes that cannot be separated out. Painting and photography, documentary and abstraction, industry and algae occupy the same surface, exposed and concealed at the same time. When Williams writes of Gowin that his contradictory earth is both abstraction and homeland, she optimistically goes on to suggest that in the aerial views '[w]hat has remained hidden is exposed. What is largely unseen, therefore unknown, is revealed. What we thought benign becomes malignant. Our perception of the world shifts' (2002: 126). This is a dream of art's emancipatory capacity which, given abstraction's duplicitous history, is too straightforward. The shift that has taken place from Gowin to Maisel – from austere black and white to violent colour; from cool elegance to garish opacity – is a move from the possibility of exposure to the recognition that it is the condition of hiddenness that is the only thing visible in the photographs. Maisel's lake is terminal; as he explains, having no natural outlets, the 'claustrophobic, no-exit, existentialist aspect of this fact sparked my curiosity'. Furthermore, to return again to the terms of the title, 'the word "mirage" seems to describe the entire hallucinatory quality of the expanse of the Great Salt Lake, the unflinching light that illuminates it and that is reflected from its surface, and the manner in which this body of work questions the nature of sight and perception' (Maisel). There is no exposure here but there is illumination. The light is unflinching but not in a revealing way; it is hallucinatory, challenging the certainties of sight and perception.

A common experience of looking at the duck–rabbit drawing is to just see one or the other: to see the duck and not consider the possibility of the rabbit. Wittgenstein uses the term 'aspect-dawning' to describe the experience of seeing the rabbit for the first time after previously only seeing the duck. While the drawing has not changed, something is different. As Malcolm Bull explains, '[a]spect-dawning is not a matter of reclassification, but of the direct apprehension of something that did not appear to be there before' (1999: 187). The rabbit was there all along but not

Figure 4.1 David Maisel, *Terminal Mirage 9* (2003). C-Print. © David Maisel/INSTITUTE.

perceived; it was not, as Bull writes, 'insensible' because 'in seeing the duck you are necessarily seeing all that is to be sensed of the rabbit'. Rather, the rabbit is hidden because it is imperceptible. When we see the duck, writes Bull, 'the rabbit is disguised, and when we see the rabbit the duck is disguised'. Once we know there is a duck and a rabbit in the drawing, while we cannot see both at once we can 'oscillate between the various possibilities more or less at will' in order to perceive each in turn (22). When we are looking at the duck or the rabbit, then, something is disguised but nothing is concealed.

The aerial photograph is both non-figurative and figurative; it is abstraction and homeland, surface and depth, concealing and exposing, art and information. Resistance to the perception of

aerial photographs as abstractions reveals a desire to maintain continuous aspect-perception, to hold the meaning of the image in place unambiguously. By visually recalling abstract painting, though, the aerial photograph opens up another aspect: the aspect of art with its history of non-figurative resistance to the penetration of the surface. This aspect-dawning means that the photograph, while unchanged, seems to show something that did not appear to be there before. The photograph is sending simultaneous messages that cannot be apprehended at the same time; something must be disguised in order for something else to be seen. Nothing is concealed, yet there is something hidden. In the end, it is not the information the photograph contains about military sites or industrial pollution that is paramount; nor is the aesthetic impact of the abstract image the most important thing. What is most compelling here is that both aspects can be sensed even when only one can be seen at a time. It is the awareness of hiddenness that the aerial photograph makes possible by generating the oscillation of perception between the indexical and the abstract.

Virilio: Hidden but Not Concealed

For Virilio, Jackson Pollock's vertically executed 'all over' paintings signify a shift from the perspectival conventions of spatial representation focused on the horizon to an outlook 'on the NADIR starting from the aerial zenith' (2007: 36). This 'scopic turning back to face the terrestrial surface' is, in Virilio's view, a consequence of aerial technology, the 'abstract world of aeroscopy celebrated by Nadar' that will, by the end of the twentieth century, end in 'the televisual MEGALOSCOPY' of real-time satellite geo-surveillance (36–7): 'Instead of observing the line that decides between Earth and sky, you observe the surface, the "support-surface", as people once contemplated the stars in the age of popular astronomy' (37). Like Sagan turning the space camera back on to Earth, the view from above renders human life undetectable at worst and at best remote and transitory. As vision is restructured by the release from the pull of gravity enabled by aerial technology, what Virilio mourns is the loss of depth and weight, of embodied experience; instead, the collapse of space represented by the horizonless surface of the abstracted ground seen from above means that the earth is no longer a support and threatens a 'fall upwards' into the abyss (1997: 2). The relationship between vision and weight

is crucial for Virilio since it embeds sight in a material world of physical forces. The Renaissance geometers might have insisted on 'near' and 'far', but for Virilio, it is not 'vanishing lines converging on the horizon' that is the original reference point for sight but rather 'one bound up with the delicate balancing act of a universal attraction which imposes on us its gearing toward the centre of the Earth' (1–2). The weightless, depthless aerial view, then, not only sweeps aside the space of perspectival illusion but any substantive relation between vision and human being in the world.

The kind of alienation Virilio registers in his assessment of technology's destruction of space is a reminder of Emerson's nineteenth-century weariness (and wariness) toward surface as well as recalling those critics who are sceptical of aerial photography's capacity to produce abstract, and therefore evasive because non-figurative, versions of real despoliation and criminal damage. What the history of abstraction in painting and photography suggests, however, is a potentially more resistant view, where the violent flattening of depth made possible by the aerial scopic regime (the panoptical rationalisation and control of space) is held in productive critical tension with abstraction's art historical tradition of non-figuration as a means of defying modes of representation that aim to capture and command.

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